

Series & Parallel Circuits

House Wiring

PHYSICS / SCIENCE · CLASS 9/10 (O-LEVEL EQUIVALENT) · 40–45 MIN

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Learning Objectives

- Differentiate between series and parallel circuits in terms of current, voltage, and resistance.
- Calculate total resistance, current, voltage drop, and power for series and parallel combinations.
- Explain why identical devices consume more total power in parallel than in series.
- Justify why domestic (house) wiring uses a parallel arrangement.

Prior Knowledge Required

- Ohm's Law ($V = IR$).
- Basic concept of current, voltage, and resistance.

Teaching Sequence

Stage	Time	Details
Recap	5 min	Quick oral review: define current, voltage, resistance using Ohm's Law.
Concept Development	15 min	Present the series vs. parallel diagram (Figure 1) and derive R_{total} for each case on the board. Work through the two-bulb example: show that $P_{\text{parallel}} = 4 \times P_{\text{series}}$ for identical bulbs at the same supply voltage.
Real-World Link — House Wiring	10 min	Show Figure 2 (house wiring diagram). Ask: 'Why doesn't your fridge get dimmer when you turn on a light?' Explain the 4 reasons: equal voltage to every appliance, independent switching, appliances draw only the current they need, individual fuses for safety.
Guided Practice	10 min	Students attempt Q1–Q3 of the worksheet in pairs; teacher circulates.
Wrap-up	5 min	Cold-call 2–3 students to justify, in their own words, why house wiring is parallel.

Key Formulas

Quantity	Series	Parallel
Current	I same throughout	I splits: $I = I_1 + I_2$
Voltage	V splits: $V = V_1 + V_2$	V same across branches
Resistance	$R_{\text{total}} = R_1 + R_2$	$1/R_{\text{total}} = 1/R_1 + 1/R_2$

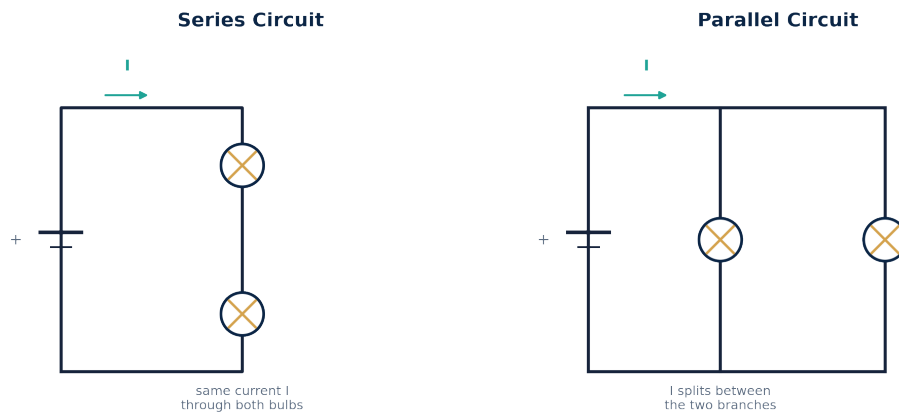


Figure 1. Series vs. parallel — two identical bulbs.

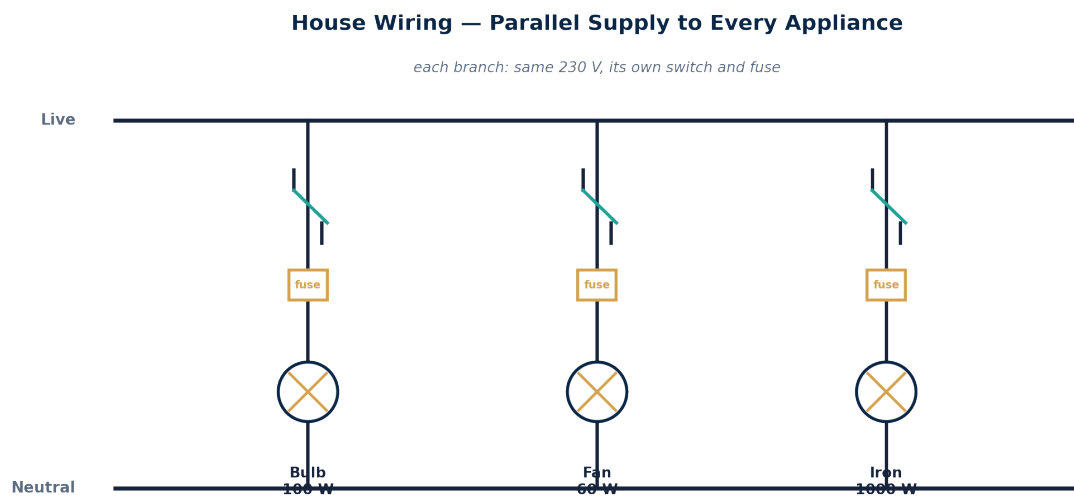


Figure 2. Parallel wiring in a house (simplified).

Assessment

Worksheet questions (attached) — collect for marking or use as homework.

Worksheet

Series & Parallel Circuits, House Wiring

Name: _____ Class: _____

Section A — Conceptual Questions (2 marks each)

- 1 State one key difference between series and parallel circuits in terms of current.
- 2 Why does removing one bulb in a series circuit switch off all the other bulbs, but not in a parallel circuit?
- 3 State two reasons why household electrical wiring is always done in parallel, not series.

Section B — Numerical Problems (5 marks each)

- 4 Two resistors of 4 ohms and 6 ohms are connected in series to a 12 V battery. Calculate: (a) total resistance, (b) current in the circuit, (c) voltage across each resistor.
- 5 The same two resistors (4 ohms and 6 ohms) are now connected in parallel to the same 12 V battery. Calculate: (a) total resistance, (b) current through each resistor, (c) total current drawn from the battery.
- 6 Two identical bulbs, each of resistance 10 ohms, are connected to a 20 V supply. Calculate the total power consumed when they are connected: (a) in series, (b) in parallel. Show that the parallel case consumes 4 times more power.
- 7 A house circuit has three appliances — a 100 W bulb, a 60 W fan, and a 1000 W iron — all connected in parallel to a 230 V supply. Explain, using the concept of parallel circuits, why each appliance receives the full 230 V regardless of the others being switched on or off.

Section C — Extended Response (5 marks)

- 8 Using the ideas of voltage division, current division, and safety, write a short paragraph (4–6 sentences) explaining why parallel wiring is essential for a modern household.

Answer key available separately for teacher use.